

Practical Week 2

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Processing triggers for libc-bin (2.43-2ubuntu2.3)
j_rakshitha@RAKSHITHA:~$ gcc command_executor.c -o command_executor
j_rakshitha@RAKSHITHA:~$ ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 2387
Child PID : 2388

Child Process
Child PID : 2388
Parent PID : 2387
Executing command: ls
OSSP command_executor command_executor.c
Child process completed.
j_rakshitha@RAKSHITHA:~$ |
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j_rakshitha@RAKSHITHA:~$ ps
  PID TTY          TIME CMD
   374 pts/0    00:00:00 bash
   940 pts/0    00:00:00 ps
j_rakshitha@RAKSHITHA:~$ top
top - 05:39:05 up 52 min,  1 user,  load average: 0.05, 0.03, 0.00
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7794.6 total,  7194.5 free,   539.3 used,   211.8 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7255.3 avail Mem

   PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24140 15412 11604 S   0.0   0.2   0:00.84 systemd
    2 root        20   0    3180   2204  2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180   2040  1940 S   0.0   0.0   0:00.00 init
   47 root        19  -1   50420 16668 15392 S   0.0   0.2   0:00.24 systemd-journal
   82 systemd+    20   0   22540 14624 12004 S   0.0   0.2   0:00.09 systemd-resolve
   89 root        20   0   35376 12364  9252 S   0.0   0.2   0:01.35 systemd-udev
  108 root        20   0   2888   1952  1820 S   0.0   0.0   0:00.05 chronyd-starter
  116 root        20   0    4472    312  2884 S   0.0   0.0   0:00.01 cron
  121 message+    20   0    8816   5412  4724 S   0.0   0.1   0:00.07 dbus-daemon
  156 root        20   0   44092 29708 14564 S   0.0   0.4   0:00.26 networkd-dispat
  162 root        20   0   18436   9404  8156 S   0.0   0.1   0:00.10 systemd-logind
  187 syslog      20   0  220548   5920  4632 S   0.0   0.1   0:00.09 rsyslogd
  211 root        20   0    5492   2764  2576 S   0.0   0.0   0:00.00agetty
  232 _chrony      20   0   24248 11684  7112 S   0.0   0.1   0:00.10 chronyd
  252 _chrony      20   0   12092   2456  1816 S   0.0   0.0   0:00.01 chronyd
  275 root        20   0  123456 32624 18016 S   0.0   0.4   0:00.20 unattended-upgr
  367 root        20   0    3184   1108    980 S   0.0   0.0   0:00.00 SessionLeader
  368 root        20   0    3200   1248   1108 S   0.0   0.0   0:00.01 Relay(374)
  374 j_raksh+    20   0    6268   5536  3864 S   0.0   0.1   0:00.09 bash
  375 root        20   0    8644   5524  4684 S   0.0   0.1   0:00.01 login
  421 j_raksh+    20   0   22508 13444 10984 S   0.0   0.2   0:00.17 systemd
  423 j_raksh+    20   0   22908   4120  2120 S   0.0   0.1   0:00.00 (sd-pam)
  451 j_raksh+    20   0    6136   5464  3912 S   0.0   0.1   0:00.04 bash
  952 j_raksh+    20   0    8172   6012  3900 R   0.0   0.1   0:00.02 top
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j_rakshitha@RAKSHITHA:~$ uname -a
Linux RAKSHITHA 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
j_rakshitha@RAKSHITHA:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:        0-11
Vendor ID:                   GenuineIntel
Model name:                  12th Gen Intel(R) Core(TM) i5-12450HX
CPU family:                   6
Model:                       151
Thread(s) per core:         2
Core(s) per socket:         6
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5376.00
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni unip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                          288 KiB (6 instances)
L1i:                          192 KiB (6 instances)
L2:                           7.5 MiB (6 instances)
L3:                           12 MiB (1 instance)
NUMA:
NUMA node(s):                 1
NUMA node0 CPU(s):           0-11
Vulnerabilities:
Gather data sampling:         Not affected
Ghostwrite:                   Not affected
Indirect target selection:    Not affected
Itlb multihit:                Not affected
L1tf:                         Not affected
Mds:                          Not affected
Meltdown:                     Not affected
Mmio stale data:              Not affected
Old microcode:                Not affected
Reg file data sampling:       Mitigation; Clear Register File
Retbleed:                     Mitigation; Enhanced IBRS
Spec rstack overflow:         Not affected
Spec store bypass:            Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:                   Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:                   Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:                         Not affected
Taa:                          Not affected
Tsx async abort:              Not affected

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